

Assignment #1

PART 1

1. Analytical Approach

Function: $y = x^2 + 5$

numerical $\rightarrow$ python code

Points: $(0,0)$, $(-4,0)$, $(-8,0)$, $(2,0)$, $(6,0)$

① distance formula

$$d(x) = \sqrt{(x-x_0)^2 + (x^2+5-y_0)^2}$$

$$D(x) = (x-x_0)^2 + (x^2+5-y_0)^2$$

② Derivative

$$D'(x) = 2(x-x_0) + 2(x^2+5-y_0)(2x)$$

③ Set derivative to zero

$$0 = 2(x-x_0) + 2(x^2+5-y_0)(2x)$$

$$0 = 2x - 2x_0 + 4x(x^2+5-y_0)$$

$$0 = 2x - 2x_0 + 4x^3 + 20x - 4xy_0$$

$$2x_0 = 4x^3 + 22x - 4xy_0$$

$$x_0 = 2x^3 + 11x - 2xy_0$$

Plug in $(0,0)$:

$$(0) = 2x^3 + 11x - 2x(0)$$

$$0 = 2x^3 + 11x$$

$$0 = x(2x^2 + 11)$$

$$\therefore x = 0, \quad 2x^2 + 11 = 0$$

$$x = \pm \sqrt{\frac{11}{2}}; \quad x \text{ not real}$$

$$y = x^2 + 5$$

$$y = 0^2 + 5$$

$$y = 5$$

closest point $(0,5)$

$$d_{\min} = \sqrt{0^2 + 5^2}$$
$$= 5$$

Plug in $(-4,0)$:

$$(-4) = 2x^3 + 11x$$

$$0 = 2x^3 + 11x + 4$$

$$0 = x^3 + \frac{11}{2}x + 2$$

$$\therefore x \approx -0.35547 \quad \text{Wolframalpha}$$

$$y = x^2 + 5$$

$$= (-0.35547)^2 + 5$$

$$= 5.1263579209$$

closest Point $(-0.35547, 5.12636)$

$$d_{\min} = \sqrt{(-0.35547 + 4)^2 + (5.1263579209)^2}$$
$$\approx 6.7267878446396$$

Plug in $(-8,0)$:

$$(-8) = 2x^3 + 11x$$

$$-4 = x^3 + \frac{11}{2}x$$

$$0 = x^3 + \frac{11}{2}x + 4$$

$$\therefore x \approx -0.67208 \quad \text{Wolframalpha}$$

$$y = x^2 + 5$$

$$= (-0.67208)^2 + 5$$

$$= 5.4516915264$$

closest Point $(-0.67208, 5.45169)$

$$d_{\min} = \sqrt{(-0.67208 + 8)^2 + (5.4516915264)^2}$$
$$= 10.243335005037$$

Plug in $(2,0)$:

$$2 = 2x^3 + 11x$$

$$1 = x^3 + \frac{11}{2}x$$

$$0 = x^3 + \frac{11}{2}x - 1$$

$$\therefore x \approx 0.18074 \quad \text{Wolframalpha}$$

$$y = x^2 + 5$$

$$= (0.18074)^2 + 5$$

$$= 5.0118243876$$

closest Point $(0.18074, 5.01182)$

$$d_{\min} = \sqrt{(0.18074 - 2)^2 + (5.0118243876)^2}$$
$$= 5.3311965244339$$

Plug $(6,0)$:

$$6 = 2x^3 + 11x$$

$$0 = x^3 + \frac{11}{2}x - 3$$

$$\therefore x \approx 0.51990 \quad \text{Wolframalpha}$$

$$y = x^2 + 5$$

$$= (0.51990)^2 + 5$$

$$= 5.27029601$$

$$d_{\min} = \sqrt{(0.51990 - 6)^2 + (5.27029601)^2}$$
$$= 7.6631254128169$$

2. Numerical Approaches

A. Newton Raphson Method

$$f(x) = f(x_0) + f'(x)(x - x_0) = 0$$

$$x^* = x = x_0 - \frac{f(x_0)}{f'(x_0)} = x_0 + \Delta x_0$$

Deriving the Formula

$$D(x) = (x - x_0)^2 + ((x^2 + 5) - y_0)^2$$

$$\begin{aligned} D'(x) &= \frac{d}{dx} (x - x_0)^2 + \frac{d}{dx} (x^2 + 5 - y_0)^2 \\ &= 2(x - x_0) + 2(x^2 + 5 - y_0)(2x) \\ &= 2(x - x_0) + 4x(x^2 + 5 - y_0) \\ &= 2x - 2x_0 + 4x^3 + 20x - 4y_0x \\ &= 4x^3 + 22x - 4y_0x - 2x_0 \end{aligned}$$

$$D''(x) = 12x^2 + 22 - 4y_0$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

since we want to minimize distance $(D(x)) \rightarrow$ want slope to be zero $(D'(x) = 0)$

$$x_{n+1} = x_n - \frac{D'(x_n)}{D''(x_n)}$$

$$x_{n+1} = x_n - \frac{4x_n^3 + 22x_n - 4y_0x_n - 2x_0}{12x_n^2 + 22 - 4y_0}$$

$$x_{n+1} = x_n - \frac{4x_n^3 + 22x_n - 2x_0}{12x_n^2 + 22}$$

All points have $y_0 = 0$

$$x_{n+1} = x_n - \frac{2x_n^3 + 11x_n - x_0}{6x_n^2 + 11}$$

Plug in $(0, 0)$: $x_0 = 0$

$$\begin{aligned} x_1 &= 0 - \frac{2(0)^3 + 11(0) - 0}{6(0)^2 + 11} \\ &= 0 \end{aligned}$$

$$|x_1 - x_0| = 0$$

$$\begin{aligned} y &= x^2 + 5 \\ &= 0^2 + 5 \\ &= 5 \end{aligned}$$

Closest Point $(0, 5)$

$$\begin{aligned} d_{\min} &= \sqrt{0^2 + 5^2} \\ &= 5 \end{aligned}$$

Plug in $(-4, 0)$: $x_0 = -4$

$$x_1 = 0 - \frac{2(0)^3 + 11(0) - (-4)}{6(0)^2 + 11}$$

$$x_1 = -\frac{4}{11}$$

$$x_2 = -\frac{4}{11} - \frac{2\left(-\frac{4}{11}\right)^3 + 11\left(-\frac{4}{11}\right) + 4}{6\left(-\frac{4}{11}\right)^2 + 11}$$

$$= -0.35273068078317 = -\frac{4140}{11737}$$

$$x_3 = -\frac{4140}{11737} - \frac{2\left(-\frac{4140}{11737}\right)^3 + 11\left(-\frac{4140}{11737}\right) + 4}{6\left(-\frac{4140}{11737}\right)^2 + 11}$$

$$= -0.35635417844$$

$$\begin{aligned} y &= (-0.35635417844)^2 + 5 \\ &= 5.1269883004916 \end{aligned}$$

$$\begin{aligned} d_{\min} &= \sqrt{(-0.35635)^2 + (5.126988)^2} \\ &= 6.2898460956012 \end{aligned}$$

Closest Point $(-0.3563, 5.1267)$

Plug in $(-8,0)$: $x_0=0$

$$x_1 = 0 - \frac{2(0)^3 + 11(0) + 8}{6(0)^2 + 11}$$

$$= -\frac{8}{11}$$

$$= -0.72$$

$$x_2 = -0.72 - \frac{2(-0.72)^3 + 11(-0.72) + 8}{6(-0.72)^2 + 11}$$

$$\doteq -0.6729923138$$

$$x_3 = -0.673 - \frac{2(-0.673)^3 + 11(-0.673) + 8}{6(-0.673)^2 + 11}$$

$$\doteq -0.6720783727$$

$$y = x^2 + 5$$

$$y \doteq x_3^2 + 5$$

$$= (-0.6721)^2 + 5$$

$$= 5.451689$$

$$d_{\min} = \sqrt{(-0.6721 + 8)^2 + (5.451689)^2}$$
$$= 9.133419514$$

closest Point $(-0.67208, 5.4517)$

Plug in $(2,0)$:

$$x_1 = 0 - \frac{2(0) + 11(0) - 2}{6(0) + 11}$$

$$= -\frac{2}{11}$$

$$\doteq -0.18$$

$$x_2 = -\frac{2}{11} - \frac{2(-\frac{2}{11})^3 + 11(-\frac{2}{11}) - 2}{6(-\frac{2}{11})^2 + 11}$$

$$\doteq 0.18074471653$$

$$x_3 = 0.1807 - \frac{2(0.1807)^3 + 11(0.1807) - 2}{6(0.1807)^2 + 11}$$

$$\doteq 0.1807446045$$

$$y = x^2 + 5$$

$$= x_3^2 + 5$$

$$= (0.18074)^2 + 5$$

$$= 5.032668612$$

$$d_{\min} = \sqrt{(0.18074 - 2)^2 + (5.03266)^2}$$
$$\doteq 5.351396$$

closest Point $(0.18074, 5.0327)$

Plug in $(6,0)$:

$$x_1 = (0) - \frac{2(0) + 11(0) - 6}{6(0) + 11}$$

$$= -\frac{6}{11}$$

$$\doteq -0.54$$

$$x_2 = -\frac{6}{11} - \frac{2(-\frac{6}{11})^3 + 11(-\frac{6}{11}) - 6}{6(-\frac{6}{11})^2 + 11}$$

$$= 0.520068157269$$

$$x_3 = 0.52 - \frac{2(0.52)^3 + 11(0.52) - 6}{6(0.52)^2 + 11}$$

$$= 0.5199036677$$

$$y = x^2 + 5$$

$$= x_3^2 + 5$$

$$= (0.5199036677)^2 + 5$$

$$= 5.270299824$$

$$d_{\min} \doteq \sqrt{(0.5199036677 - 6)^2 + (5.2702998)^2}$$
$$\doteq 7.603125413$$

closest Point $(0.51990, 5.2703)$

B. Golden Section Search

Equation to minimize: $D(x) = (x - x_0)^2 + (x^2 + 5 - y_0)^2$ all points have $y_0 = 0$

$$D(x) = (x - x_0)^2 + (x^2 + 5)^2$$

Choose $[a, b] = [-2, 2]$

Let u, v represent points on the parabola within the interval $a < u < v < b$.

Plug in $(0, 0)$:

$$D'(x) = 2(x - x_0) + 4x(x^2 + 5)$$

$$\text{Golden ratio: } \phi = \frac{1 + \sqrt{5}}{2} \approx 1.618034$$

$$\frac{1}{\phi} = \phi - 1 \approx 0.618034$$

$$c = 1 - \frac{1}{\phi}$$

$$c = 1 - (\phi - 1)$$

$$c = 2 - \phi \approx 0.381966$$

$$\text{Interior Points: } u = a + c(b - a) \rightarrow u = (-2) + (0.381966)(2 - (-2)) \doteq -0.472136$$

$$v = b - c(b - a) \rightarrow v = (2) + (0.381966)(2 - (-2)) \doteq 0.472136$$

Point $(0, 0)$:

$$\textcircled{1} \quad D(u) = (-0.472136)^2 + ((-0.472136)^2 + 5)^2 \\ \doteq 27.501726$$

$$D(v) = (0.472136)^2 + ((0.472136)^2 + 5)^2 \\ \doteq 27.501726$$

$$D(u) = D(v)$$

$$\Rightarrow a = u$$

$$\Rightarrow [a, b] = [-0.472136, 2]$$

$$\textcircled{2} \quad u = -0.472136 + c(2 - (-0.472136)) \\ \doteq 0.472136$$

$$v = 2 - c(2 - (-0.472136)) \\ \doteq 1.055728$$

$$D(u) \approx 27.501726 \quad (\text{same as prev } D(v))$$

$$D(v) = (1.055728)^2 + ((1.055728)^2 + 5)^2 \\ \approx 38.502428$$

$$D(u) < D(v)$$

$$\Rightarrow b = v$$

$$\Rightarrow [a, b] = [-0.472136, 1.055728]$$

Point $(-4, 0)$:

$$\textcircled{1} \quad D(u) = (-0.472136 - (-4))^2 + ((-0.472136)^2 + 5)^2 \\ \doteq 39.7246$$

$$D(v) = (0.472136 - (-4))^2 + ((0.472136)^2 + 5)^2 \\ \doteq 47.2788$$

$$D(u) < D(v)$$

$$\Rightarrow b = v$$

$$\Rightarrow [a, b] = [-2, 0.472136]$$

$$\textcircled{2} \quad u = 0.472136 - r(0.472136 - (-2)) \\ \doteq -1.0557$$

$$v = -2 + r(0.472136 - (-2)) \\ \doteq -0.472136$$

$$D(u) = (-1.0557 - (-4))^2 + ((-1.0557)^2 + 5)^2 \\ \doteq 46.056$$

$$D(v) = (-0.472136 - (-4))^2 + ((-0.472136)^2 + 5)^2 \\ \doteq 39.725$$

$$D(u) > D(v)$$

$$\Rightarrow a = u$$

$$\Rightarrow [a, b] = [-1.0557, 0.472136]$$

Point $(-8,0)$:

$$\textcircled{1} D(u) = (-0.472136 - (-8))^2 + ((-0.472136)^2 + 5)^2$$

$$\doteq 83.9476$$

$$D(v) = (0.472136 - (-8))^2 + ((0.472136)^2 + 5)^2$$

$$\doteq 99.0659$$

$$D(u) < D(v)$$

$$\Rightarrow b = v$$

$$\Rightarrow [a, b] = [-2, 0.472136]$$

$$\textcircled{1} u = 0.472136 - r(0.472136 - (-2))$$

$$\doteq -1.05573$$

$$v = -2 + r(0.472136 - (-2))$$

$$\doteq -0.472136$$

$$D(u) = (-1.05573 - (-8))^2 + ((-1.05573)^2 + 5)^2$$

$$\doteq 85.6108$$

$$D(v) = (-0.472136 - (-8))^2 + ((-0.472136)^2 + 5)^2$$

$$\doteq 83.9476$$

$$D(u) > D(v)$$

$$\Rightarrow a = u$$

$$\Rightarrow [a, b] = [-1.05573, 0.472136]$$

Point $(2,0)$:

$$\textcircled{1} D(u) = (-0.472136 - (2))^2 + ((-0.472136)^2 + 5)^2$$

$$\doteq 33.3903$$

$$D(v) = (0.472136 - (2))^2 + ((0.472136)^2 + 5)^2$$

$$\doteq 29.6132$$

$$D(u) > D(v)$$

$$\Rightarrow a = u$$

$$\Rightarrow [a, b] = [-0.472136, 2]$$

$$\textcircled{2} u = 2 + r(2 - (-0.472136))$$

$$\doteq 0.472136$$

$$v = -0.472136 + r(2 - (-0.472136))$$

$$\doteq 1.05573$$

$$D(u) = (0.472136 - (2))^2 + ((0.472136)^2 + 5)^2$$

$$\doteq 29.6132$$

$$D(v) = (1.05573 - (2))^2 + ((1.05573)^2 + 5)^2$$

$$\doteq 38.2795$$

$$D(u) < D(v)$$

$$\Rightarrow b = v$$

$$\Rightarrow [a, b] = [-0.472136, 1.05573]$$

Point $(6,0)$:

$$\textcircled{1} D(u) = (-0.472136 - (6))^2 + ((-0.472136)^2 + 5)^2$$

$$\doteq 69.1674$$

$$D(v) = (0.472136 - (6))^2 + ((0.472136)^2 + 5)^2$$

$$\doteq 57.8361$$

$$D(u) > D(v)$$

$$\Rightarrow a = u$$

$$\Rightarrow [a, b] = [-0.472136, 2]$$

$$\textcircled{2} u = 2 - r(2 - (-0.472136))$$

$$\doteq 0.472136$$

$$v = -0.472136 - r(2 - (-0.472136))$$

$$\doteq 1.05573$$

$$D(u) = (0.472136 - (6))^2 + ((0.472136)^2 + 5)^2$$

$$\doteq 57.8361$$

$$D(v) = (1.05573 - (6))^2 + ((1.05573)^2 + 5)^2$$

$$\doteq 61.8337$$

$$D(u) < D(v)$$

$$\Rightarrow b = v$$

$$\Rightarrow [a, b] = [-0.472136, 1.05573]$$

PART 2

Fitting a Line & Parabola

Points: (0, 0.5), (2, 3.5), (1, 1.5), (3, 7.5)

$$\text{MSE: } E = \frac{1}{4} \sum_{i=1}^4 (\hat{y}_i - y_i)^2$$

1. Line

$$\hat{y} = mx + b$$

$$E(m, b) = \frac{1}{4} [(b - 0.5)^2 + (2m + b - 3.5)^2 + (m + b - 1.5)^2 + (3m + b - 7.5)^2]$$

Analytical Solution

$$\frac{\partial E}{\partial m} = \frac{1}{2} \sum_i x_i (mx_i + b - y_i) = 0$$

$$0 = \frac{1}{2} (14m + 6b - 31)$$

$$\textcircled{1} 31 = 14m + 6b$$

$$\frac{\partial E}{\partial b} = \frac{1}{2} \sum_i (mx_i + b - y_i) = 0$$

$$0 = \frac{1}{2} (6m + 4b - 13)$$

$$\textcircled{2} 13 = 6m + 4b$$

$$2 \times \textcircled{1} - 3 \times \textcircled{2}:$$

$$2(14m + 6b) - 3(6m + 4b) = 62 - 39$$

$$10m = 23$$

$$m = 2.3$$

$$E_{\min} = \frac{1}{4} [(-0.2 - 0.5)^2 + (4.40 - 3.5)^2 + (2.10 - 1.5)^2 + (6.70 - 7.5)^2]$$
$$= 0.5750$$

$$b = \frac{13 - 6(2.3)}{4} = -0.2$$

$$\hat{y} = 2.3x - 0.2$$

Newton-Raphson: single Parameter update

$$\theta_j^{\text{new}} = \theta_j - \frac{\frac{\partial E}{\partial \theta_j}}{\frac{\partial^2 E}{\partial \theta_j^2}}$$

$$m_{k+1} = \frac{31 - 6b_k}{14}$$

$$b_{k+1} = \frac{13 - 6m_{k+1}}{4}$$

Starting w/ all parameters at zero

$$\textcircled{1} m^{(1)} = \frac{31 - 6(0)}{14}$$

$$\hat{=} 2.2143$$

$$b^{(1)} = \frac{13 - 6(2.2143)}{4}$$

$$\hat{=} -0.071429$$

$$E^{(1)} = \frac{1}{4} [(-0.07143 - 0.5)^2 + (4.357 - 3.5)^2 + (2.143 - 1.5)^2 + (6.571 - 7.5)^2]$$

$$\hat{=} 0.5892$$

$$\textcircled{2} m^{(2)} = \frac{31 - 6(-0.07143)}{14}$$

$$\hat{=} 2.2449$$

$$b^{(2)} = \frac{13 - 6(2.2449)}{4}$$

$$\hat{=} -0.11735$$

$$E^{(2)} = \frac{1}{4} [(-0.11735 - 0.5)^2 + (4.3724 - 3.5)^2 + (2.1276 - 1.5)^2 + (6.6173 - 7.5)^2]$$

$$\hat{=} 0.5788$$

2. Parabola Points: (0, 0.5), (2, 3.5), (1, 1.5), (3, 7.5)

$$\hat{y} = ax^2 + bx + c$$

MSE:

$$E(a, b, c) = \frac{1}{4} [(c - 0.5)^2 + (4a + 2b + c - 3.5)^2 + (a + b + c - 1.5)^2 + (9a + 3b + c - 7.5)^2]$$

Analytical Solution

$$\frac{\partial E}{\partial a} = \frac{1}{2} \sum_i x_i^2 (ax_i^2 + bx_i + c - y_i) = 0$$

$$0 = \frac{1}{2} (98a + 36b + 14c - 83) \quad (1)$$

$$\frac{\partial E}{\partial b} = \frac{1}{2} \sum_i x_i (ax_i^2 + bx_i + c - y_i) = 0$$

$$0 = \frac{1}{2} (36a + 14b + 6c - 31) \quad (2)$$

$$\frac{\partial E}{\partial c} = \frac{1}{2} \sum_i (ax_i^2 + bx_i + c - y_i) = 0$$

$$0 = \frac{1}{2} (14a + 6b + 4c - 13) \quad (3)$$

$$4 \times (1) - 14 \times (3): \quad 0 = 2(98a + 36b + 14c - 83) - 7(14a + 6b + 4c - 13)$$

$$75 = 98a + 30b$$

$$4 \times (2) - 6 \times (3): \quad 0 = 2(36a + 14b + 6c - 31) - 3(14a + 6b + 4c - 13)$$

$$23 = 30a + 10b$$

$$(98a + 30b) - 3(30a + 10b) = 75 - 69$$

$$\Rightarrow 8a = 6$$

$$a = 0.75$$

$$b = \frac{23 - 30(0.75)}{10} = 0.05$$

$$c = \frac{13 - 14(0.75) - 6(0.05)}{4} = 0.55$$

$$\hat{y} = 0.75x^2 + 0.05x + 0.55$$

$$E_{\min} = \frac{1}{4} [(0.55 - 0.5)^2 + (3.65 - 3.5)^2 + (1.35 - 1.5)^2 + (7.45 - 7.5)^2]$$

$$= 0.0125$$

Newton-Raphson: single parameter updates

$$y = ax^2 + bx + c$$

$$E(a, b, c) = \frac{1}{4} \sum_{i=1}^4 (ax_i^2 + bx_i + c - y_i)^2$$

$$\sum x_i = 6 \quad \sum y_i = 13$$

$$\sum x_i^2 = 14 \quad \sum x_i y_i = 31$$

$$\sum x_i^3 = 36 \quad \sum x_i^2 y_i = 83$$

$$\sum x_i^4 = 98$$

$$\theta_j^{\text{new}} = \theta_j - \frac{\frac{\partial E}{\partial \theta_j}}{\frac{\partial^2 E}{\partial \theta_j^2}}$$

$$\frac{\partial E}{\partial a} = \frac{1}{2} \sum x_i^2 (ax_i^2 + bx_i + c - y_i)$$

$$= \frac{1}{2} (98a + 36b + 14c - 83)$$

$$\frac{\partial^2 E}{\partial a^2} = \frac{98}{2} = 49$$

$$a_{k+1} = a_k - \frac{\frac{1}{2} (98a_k + 36b_k + 14c_k - 83)}{49}$$

$$= a_k - \frac{98a_k + 36b_k + 14c_k - 83}{98}$$

$$a_{k+1} = \frac{83 - 36b_k - 14c_k}{98}$$

$$\frac{\partial E}{\partial b} = \frac{1}{2} \sum x_i (a x_i^2 + b x_i + c - y_i)$$

$$= \frac{1}{2} (a \sum x_i^3 + b \sum x_i^2 + c \sum x_i - \sum x_i y_i)$$

$$= \frac{1}{2} (36a + 14b + 6c - 31)$$

$$\frac{\partial^2 E}{\partial b^2} = \frac{14}{2} = 7$$

$$b_{k+1} = b_k - \frac{\frac{1}{2} (36a_{k+1} + 14b_k + 6c_k - 31)}{7}$$

$$= b_k - \frac{36a_{k+1} + 14b_k + 6c_k - 31}{14}$$

$$b_{k+1} = \frac{31 - 36a_{k+1} - 6c_k}{14}$$

$$\frac{\partial E}{\partial c} = \frac{1}{2} \sum (a x_i^2 + b x_i + c - y_i)$$

$$= \frac{1}{2} (a \sum x_i^2 + b \sum x_i + 4c - \sum y_i)$$

$$= \frac{1}{2} (14a + 6b + 4c - 13)$$

$$\frac{\partial^2 E}{\partial c^2} = \frac{4}{2} = 2$$

$$c_{k+1} = c_k - \frac{\frac{1}{2} (14a_{k+1} + 6b_{k+1} + 4c_k - 13)}{2}$$

$$= \frac{4c_k - 14a_{k+1} - 6b_{k+1} - 4c_k + 13}{4}$$

$$c_{k+1} = \frac{13 - 14a_{k+1} - 6b_{k+1}}{4}$$

$$\textcircled{v} a^{(1)} = \frac{23 - 36(0) - 14(0)}{98} \hat{=} 0.234694$$

$$b^{(1)} = \frac{31 - 36(0.234694) - 6(0)}{14} \hat{=} 0.036443$$

$$c^{(1)} = \frac{13 - 14(0.234694) - 6(0.036443)}{4} \hat{=} 0.23105$$

$$E^{(1)} = \frac{1}{4} [(0.231049563 - 0.5)^2 + (3.691690762 - 3.5)^2 + (1.114431487 - 1.5)^2 + (7.962727988 - 7.5)^2]$$

$$\hat{=} 0.117988147$$